1 Worksheet

Question 1.1. Let $E \in \mathcal{E}$ satisfy $m(E) > 0$. Show that there is a bounded set $A \subseteq E$ such that $m(A) > 0$.

Proof. Consider the increasing collection $\{A_n := (-n, n) \cap E\}_{n=1}^{\infty}$ of measurable sets. By continuity of measure, we have

$$\lim_{n\to\infty} m(A_n) = m\left(\bigcup_{n=1}^{\infty} A_n\right) = m(E) > 0$$

Since the limit of $m(A_n) > 0$, we must have that $m(A_n) > 0$ for at least one n. Hence $A := (-n, n) \cap E \subseteq E$ is a bounded set with $m(A) > 0$. $\square$

Question 1.2. Show that every open set $O \subset \mathbb{R}$ is a countable union of disjoint intervals.

Proof. For each $x \in O$, there is some $\varepsilon > 0$ such that $(x - \varepsilon_x, x + \varepsilon_x) \subseteq O$. Note that if $\delta > 0$ is such that for all $0 < \varepsilon < \delta$, $(x - \varepsilon, x + \varepsilon) \subseteq O$, then $(x - \delta, x + \delta) = \bigcup_{0<\varepsilon<\delta}(x - \varepsilon, x + \varepsilon) \subseteq O$. We thus let $\delta_x = \sup\{\varepsilon > 0 : (x - \varepsilon, x + \varepsilon) \subseteq O\} > 0$, which is finite since $O \neq \mathbb{R}$ and satisfies $U_x := (x - \delta_x, x + \delta_x) \subseteq O$.

By construction of the δ_x's, we have $\bigcup_{x\in O\cap\mathbb{Q}} U_x$, a countable union of disjoint intervals, is a subset of O. For the reverse equality, for any point $y \in O$, we take $\varepsilon > 0$ for which $(y - 2\varepsilon, y + 2\varepsilon) \subseteq O$, and we take some rational point $x \in (y - \varepsilon, y + \varepsilon) \subseteq O$. Note that then $y \in (x - \varepsilon, x + \varepsilon) \subseteq (y - 2\varepsilon, y + 2\varepsilon) \subseteq O$, so by construction, $y \in (x - \varepsilon, x + \varepsilon) \subseteq U_x$. Therefore, $O \subseteq \bigcup_{x\in O\cap\mathbb{Q}} U_x$, which proves that O is a countable union of disjoint open intervals. $\square$

Question 1.3. Let $E \in \mathcal{E}$ be Lebesgue measurable with $m(E) > 0$. Let $0 < \alpha < 1$. Show that there is a bounded interval I so that

$$m(E \cap I) \geq \alpha m(I)$$

Proof. For any $A \subseteq E$, we have $m(E \cap I) \geq m(A \cap I)$ for any bounded interval I. By question 1, there is a bounded set A with positive measure, so we may assume that E is already bounded. Take $M > 0$ such that $E \subseteq (-M, M)$. By way of contradiction, suppose that $m(E \cap I) < \alpha m(I)$ for all bounded intervals I. For a fixed $\varepsilon > 0$, take open intervals $\{I_k\}_{k=1}^{\infty}$ that cover E and $\sum_{k=1}^{\infty} \ell(I_k) < m(E) + \varepsilon$. We then have

$$m(E) \leq \sum_{k=1}^{\infty} m(E \cap I_k) < \alpha \sum_{k=1}^{\infty} \ell(I_k) < \alpha m(E) + \alpha\varepsilon$$

Letting $\varepsilon \to 0$, we see that $m(E) \leq \alpha m(E)$, which is impossible since $m(E) \in (0, \infty)$ (recall that we took E to be bounded) and $\alpha \in (0, 1)$. Therefore, we must have some bounded interval I for which $m(E \cap I) \geq \alpha m(I)$. $\square$

Question 1.4. Let $E \in \mathcal{E}$ be Lebesgue measurable with $m(E) > 0$. Prove that the set $A = E - E := \{x - y : x \in E, y \in E\}$ contains an open interval J such that $0 \in J$.

Proof. By question 3, we take a bounded interval I such that $m(E \cap I) \geq \frac{3}{4} m(I)$. If we suppose that the result does not hold, then we can take a sequence $x_n \to 0$ such that $x_n \notin E - E$. We take n large enough such that $|x_n| < \frac{1}{4}m(I)$ and note that $x_n + E \subseteq E^c$. We then get the following:

$$\begin{aligned}
m(I) &= m(E \cap I) + m(I \cap E^c) \\
&\geq m(E \cap I) + m(I \cap (x_n + E)) \\
&= m(E \cap I) + m((I - x_n) \cap E) \\
&\geq 2m(E \cap I) - m(((I + x_n) \setminus (I - x_n)) \cap E) \\
&\geq 2m(E \cap I) - m((I + x_n) \setminus (I - x_n)) \\
&\geq 2m(E \cap I) - 2|x_n| \\
&> \frac{3}{2}m(I) - \frac{1}{2}m(I) \\
&= m(I)
\end{aligned}$$

which is a contradiction. □

Question 1.5. Let $f : \mathbb{R} \to \mathbb{R}$ be a Lebesgue measurable function such that

$$f(x + y) = f(x) + f(y)$$

Show that $f(x) = ax$ for some $a \in \mathbb{R}$. **Hint:** Apply Lusin's theorem and the previous problem.

Proof. Use Lusin's theorem to get a closed set F with $m(F^c) < 1/2$ and a continuous $g : F \to \mathbb{R}$ such that $f = g$ on F. By question 1, we can also restrict F to a compact set while keeping $m(F) > 0$. We can then apply question 4 to get an open interval $(-M, M)$ contained in $F - F$. Since F is compact, g is also uniformly continuous on F, so given an $\varepsilon > 0$, there is some $\delta > 0$ for which $|g(x) - g(y)| < \varepsilon$ for any $x, y \in F, |x - y| < \delta$. For any $h \in (-\delta, \delta)$, we take $x \in F$ for which $h = (x + h) - x \in F - F$ and get $f(h) = f(x + h) - f(x) = g(x + h) - g(x) \Rightarrow |f(h)| < \varepsilon$. This shows that f is continuous at 0. For any arbitrary $x_0 \in \mathbb{R}$, we can write $f(x + x_0) = f(x) + f(x_0)$ which shows that f is continuous at x_0, therefore f is continuous on $\mathbb{R}$. By continuity, it then suffices to only show that $f(x) = ax$ on $x \in \mathbb{Q}$.

Set $a := f(1)$. $f(0) = f(0) + f(0) \Rightarrow f(0) = 0$ and for any $x \in \mathbb{Q}$, we have $f(-x) = f(0) - f(x) = -f(x)$. For any $n \in \mathbb{N}$, we have $f(n) = \sum_{i=1}^{n} f(1) = an$, and $f(-n) = -f(n) = -an$, so $f(x) = ax$ on $\mathbb{Z}$. For any $n \in \mathbb{N}$, we have $nf(1/n) = f(1) = a \Rightarrow f(1/n) = a/n$ and for any $p, q \in \mathbb{N}$, we have $qf(p/q) = f(p) = ap \Rightarrow f(p/q) = ap/q$, so for any $x \in \mathbb{Q}, f(x) = ax$. As noted earlier, continuity of f shows that $f(x) = ax$ on $\mathbb{R}$. □

Question 1.6. Construct a non-linear $f : \mathbb{R} \to \mathbb{R}$ obeying $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. (Hint: consider $\mathbb{R}$ as a vector space with scalar field $\mathbb{Q}$)

Proof. Take a basis β for $\mathbb{R}$ as a vector space over $\mathbb{Q}$. This gives us a unique expression of any real number x as $q_1e_1 + \cdots + q_ne_n$ for $q_i \in \mathbb{Q} \setminus \{0\}, e_i \in \beta$. This lets us define f by $q_1e_1 + \cdots + q_ne_n \mapsto q_1 + \cdots + q_n$. This function is obviously additive, but it cannot be continuous on $\mathbb{R}$, since its image is the disconnected set $\mathbb{Q}$. Hence it is not linear. □

2 Royden Chapter 3

Question 2.1 (Royden 3.4)**.** Suppose f is a real-valued function on $\mathbb{R}$ such that $f^{-1}(c)$ is measurable for each number c. Is f necessarily measurable?

Proof. No. Let E be any non-measurable set and define $f(x) = e^x(2\chi_E(x) - 1)$. Notice that $f^{-1}((0, \infty)) = E$ and $f^{-1}((-\infty, 0)) = E^c$, so we already see that f is not measurable. Furthermore, if $f(x) = f(y)$, we have either $x, y \in E$ or $x, y \in E^c$, so in either case $e^x = e^y \Rightarrow x = y$, and hence $f^{-1}(c)$ for any $c \in \mathbb{R}$ is either empty or a singleton, hence measurable. □

Question 2.2 (Royden 3.5)**.** Suppose the function f is defined on a measurable set E and has the property that $\{x \in E \mid f(x) > c\}$ is measurable for each rational number c. Is f necessarily measurable?

Proof. It suffices to show that $f^{-1}((c,\infty))$ is measurable for any real c. By density of $\mathbb{Q}$, we can take a rational sequence $c_n > c$ such that $c_n \to c$. We then see that $\bigcup_{n=1}^{\infty}(c_n,\infty) = (c,\infty)$ and hence $f^{-1}((c,\infty)) = \bigcup_{n=1}^{\infty} f^{-1}((c_n,\infty))$ is measurable as a countable union of measurable sets. □

Question 2.3 (Royden 3.6)**.** Let f be a function with measurable domain D. Show that f is measurable if and only if the function g defined on $\mathbb{R}$ by

$$g(x) = \begin{cases} f(x) & \text{for } x \in D \\ 0 & \text{for } x \notin D \end{cases}$$

is measurable.

Proof. Note that for $c \geq 0$, we have $g^{-1}((c,\infty)) = f^{-1}(c,\infty)$ and for $c < 0$, $g^{-1}((c,\infty)) = D^c \cup f^{-1}((c,\infty))$ $\iff$ $f^{-1}((c,\infty)) = g^{-1}((c,\infty)) \cap D$. The result follows after observing that D, D^c are measurable. □

Question 2.4 (Royden 3.13)**.** A real-valued measurable function is said to be *semisimple* provided it takes only a countable number of values. Let f be any measurable function on E. Show that there is a sequence of semisimple functions $\{f_n\}$ on E that converges to f uniformly on E.

Proof. We note that if f is extended real-valued then this is not possible since $|f(x) - \varphi(x)| = \infty$ for any x such that $f(x) = \pm\infty$ and any real-valued φ. We thus assume that f is real-valued. Also by writing f as a difference of its positive and negative parts (which are measurable), we may assume that f is nonnegative.

Define E_n to be the measurable set $E_n := f^{-1}([n, n+1))$, which are mutually disjoint and cover E. For each n, the restriction f_n of f to E_n is a bounded measurable function, so for any $\varepsilon > 0$, f can be approximated by simple functions $\varphi_n^\varepsilon, \psi_n^\varepsilon$ on E_n, that is $\varphi_n^\varepsilon \leq f_n \leq \psi_n^\varepsilon$ and $0 \leq \psi_n^\varepsilon - \varphi_n^\varepsilon < \varepsilon$. Also by extending $\varphi_n^\varepsilon, \psi_n^\varepsilon$ to E in the way of Royden 3.6, we get that $\varphi_n^\varepsilon, \psi_n^\varepsilon$ are simple functions on E. We define the semisimple functions $\varphi^\varepsilon = \sum_{i=0}^{\infty} \varphi_n^\varepsilon, \psi^\varepsilon = \sum_{i=0}^{\infty} \psi_n^\varepsilon$ and note that their restrictions to each E_n are precisely $\varphi_n^\varepsilon, \psi_n^\varepsilon$. This implies that $\varphi^\varepsilon \leq f \leq \psi^\varepsilon$ and $0 \leq \psi^\varepsilon - \varphi^\varepsilon < \varepsilon$ on E, and hence $0 \leq f - \varphi^\varepsilon \leq \psi^\varepsilon - \varphi^\varepsilon < \varepsilon$. The result follows by taking the sequence $\varphi^{1/n}$. □

Question 2.5 (Royden 3.14)**.** Let f be a measurable function on E that is finite a.e. on E and $m(E) < \infty$. For each $\epsilon > 0$, show that there is a measurable set F contained in E such that f is bounded on F and $m(E \setminus F) < \epsilon$.

Proof. Let E_0 be the set of measure zero on which f is infinite. Consider the increasing collection $\{F_n := f^{-1}((-n,n))\}_{n=1}^{\infty}$ of measurable sets and note that the union over it is $E \setminus E_0$. By continuity of measure, we have $m(E) = m(E \setminus E_0) = m(\bigcup_{n=1}^{\infty} F_n) = \lim_{n\to\infty} m(F_n)$. For any $\varepsilon > 0$, we have $0 \leq m(E) - m(F_n) < \varepsilon$ for large enough n, hence there is a measurable $F_n \subseteq E$ for which $m(E \setminus F_n) = m(E) - m(F_n) < \varepsilon$ and $f(F_n) \subseteq (-n,n)$ is bounded. □

Question 2.6 (Royden 3.26)**.** For the function f and the set F in the statement of Lusin's Theorem, show that the restriction of f to F is a continuous function. Must there be any points at which f, considered as a function on E, is continuous?

Proof. The result of Lusin's theorem is that $f = g$ on F for a continuous function $g : \mathbb{R} \to \mathbb{R}$. Since g is continuous on F, then so must be f. f as a function on E might not be continuous at any point however. As an example, consider the simple function $f = \chi_{\mathbb{Q}\cap[0,1]} : [0,1] \to \mathbb{R}$ which is discontinuous everywhere on $[0,1]$, but by Lusin's theorem can be restricted to a continuous function. □

Question 2.7 (Royden 3.27)**.** Show that the conclusion of Egoroff's Theorem can fail if we drop the assumption that the domain has finite measure.

Proof. ◻

Question 2.8 (Royden 3.29)**.** Prove the extension of Lusin's Theorem to the case that E has infinite measure.

Proof. ◻